lit2026 W10: Coppersmith reach for the prime-field S_5 small-root oracle

2026-10-11 · branch `lit2026/w10-s5-coppersmith` · engine `examples/lit2026_w10_coppersmith_bound.rs` @ `a21b5de50` · run `run-20261011T2118Z`

Question. Index calculus over $\mathbb{F}_p$ first beats rho at arity $m = 4$. That needs roots of $f = S_5(x_1, \ldots, x_4, z) \equiv 0 \bmod p$ with $|x_i| < p^\delta$, $\delta \approx \frac{1}{5}$ (repo `RESEARCH_PRIME_FIELD_BREAKTHROUGH_PROGRAM.md` §2.4). Do the 2024–26 automated-Coppersmith tools (Ryan ePrint 2024/1577 "cuso"; Ding et al. 2026/1027, an LP for Meers–Nowakowski asymptotics; Ding et al. 2026/423) reach that? Only the abstracts were read: the 2026/1027 PDF returned HTTP 403 (bot protection, not bypassed). The optimiser below is our own exact formulation.

Problem. f has degree $d = 2^{m-1}$ per variable. Measured support with z substituted (61-bit field): S_3 9/9, S_4 125/125, S_5 6561/6561 monomials of the box $[0, d]^m$, all of them present. Total degree md = 4, 12, 32. Checked against genuine relations and planted decompositions.

Family $\mathcal{F}$. Lattices spanned by $p^{\ell-k(\beta)} x^{\beta-k(\beta)a} f^{k(\beta)}$, $\beta \in M$, with every support inside M; a is a vertex of $\mathrm{Newt}(f)$. As $p \to \infty$ the lattice succeeds iff $\sum_M k(\beta) > \sum_M \lambda \cdot \beta$. This family contains Jochemsz–May basic and extended shifts and every monomial-set choice for one polynomial over $(p, f)^\ell$.

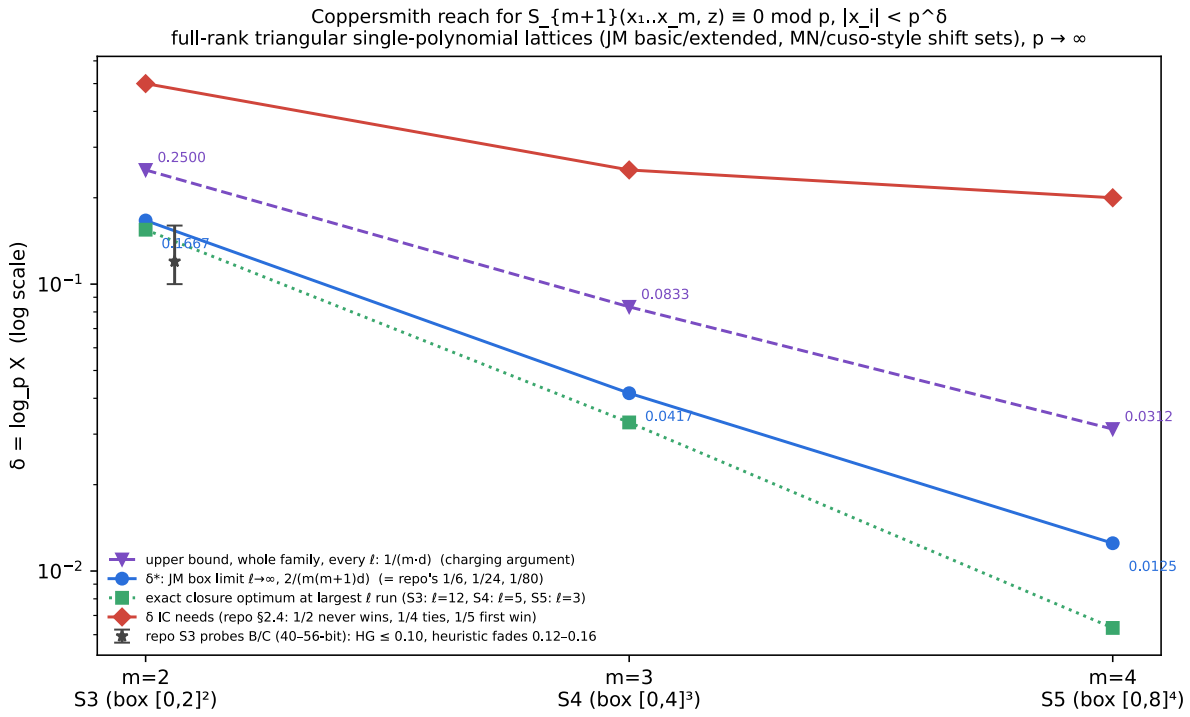
Exact optimiser. For fixed (a, ℓ) and a bounding box, the choice of (M, k) is a maximum-weight closure (totally unimodular LP). We solve it by min-cut inside Dinkelbach iteration and get an exact rational.

Ceiling. Charge the j-th unit of $k(\beta)$ to $\beta + j(c - a)$. This gives $\sum k \le \sum_M \frac{\lambda \cdot u}{\lambda \cdot c}$, so success needs $\lambda \cdot c < 1$ for the top vertex c, i.e. $\delta < \frac{1}{md}$ at every ℓ. The bound holds for the e-encoding as well, whose weighted degree is the same md.

	S_3 ($m = 2$)	S_4 ($m = 3$)	S_5 ($m = 4$)
δ^* in $\mathcal{F}$ (box limit $2/(m(m+1)d)$)	**1/6**	**1/24**	**1/80 = 0.0125**
ceiling of $\mathcal{F}$, every ℓ: $1/(md)$	1/4	1/12	1/32 = 0.03125
exact closure optimum (largest ℓ)	0.154789 ($\ell = 12$)	0.032955 ($\ell = 5$)	0.006334 ($\ell = 3$)
closure, e-encoding	0.136111 ($\ell = 3$)	0.024903 ($\ell = 2$)	0.004061 ($\ell = 1$)
δ needed (repo §2.4)	1/2	1/4 (tie)	**1/5**
repo / this run, finite lattices	HG ≤ 0.10, heuristic to 0.14–0.16	ℓ=1, dim 125: 0/36 at B = 2…15	—

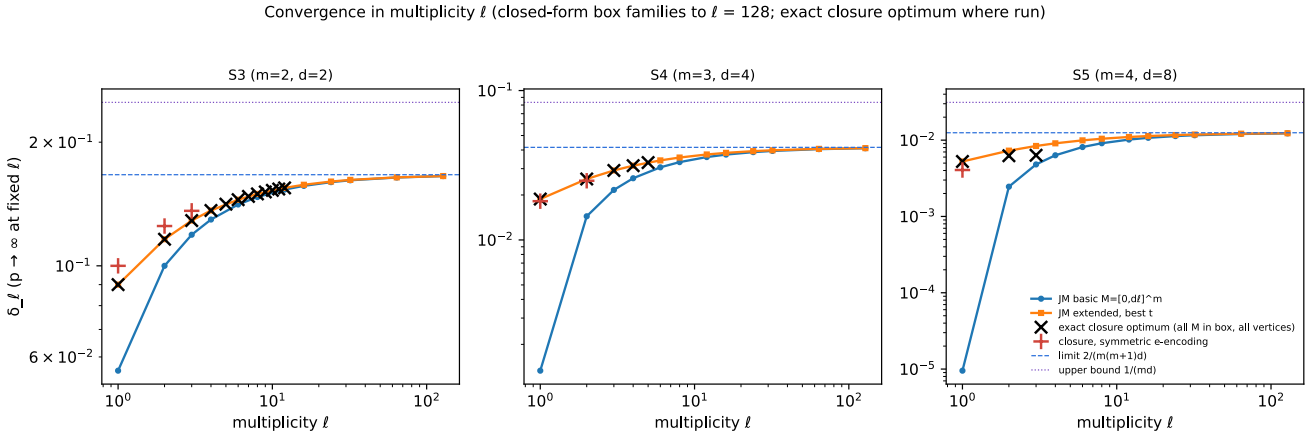
Validation. Univariate: the closure optimum equals $\ell/(d\ell + d - 1)$ in 30/30 cells and tends to $1/d$. Boneh–Durfee: the closure optimum equals the x+y-shift lattice with the best t at every $\ell \le 32$, and that lattice reaches 0.28466 at $\ell = 2000$ against the literature value 0.28475. The 0.2929 bound lies outside $\mathcal{F}$ (it needs a sublattice). JM basic gives 1/18, 1/10, 0.119 for S_3, the repo's column. In 70/70 closure cells (S_3 $\ell \le 12$, S_4 $\ell \le 5$, S_5 $\ell \le 3$, every vertex class) the optimum over all monomial sets is a box $[0, L']^m$.

Verdict. $\delta^*(S_5) = \frac{1}{80} < \frac{1}{5}$, and the ceiling of $\mathcal{F}$ is $\frac{1}{32} < \frac{1}{5}$. **The Coppersmith route to the $m = 4$ oracle is closed for full-rank triangular single-polynomial lattices over $(p, f)^\ell$**: JM basic and extended shifts, all monomial sets, all leading vertices, in both the x-encoding and the e-encoding. **Not covered:** non-full-rank sublattices and relation-based linearisation (the Boneh–Durfee 0.292 / Herrmann–May type); cuso's Gröbner-optimal combinations that cancel out-of-M monomials; multi-polynomial systems (curve equations, shared unknowns); unequal or non-interval bounds. **Correction to the repo:** the "measured 1/80" in `RESEARCH_IC_LEADS_AND_CONNECTIONS_20261010.md` is the derived JM-basic limit. Only S_3 was ever measured.



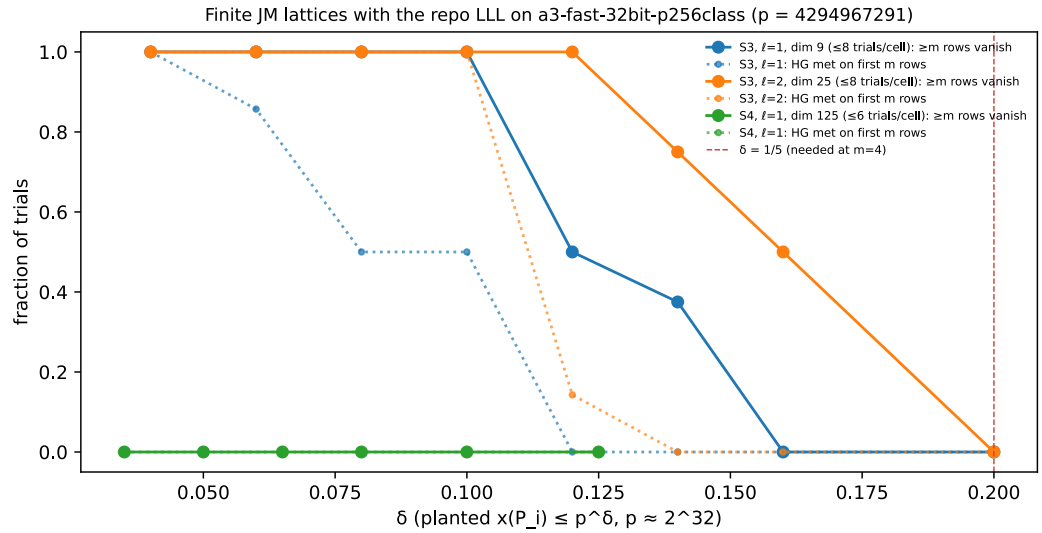
Source: research/lit2026_w10_s5_coppersmith_20261011/reports/run-20261011T2118Z/run.json (lit2026 W10, examples/lit2026_w10_coppersmith_bound.rs)

Figure 1: δ against arity. Source `reports/run-20261011T2118Z/run.json`.



Source: research/lit2026_w10_s5_coppersmith_20261011/reports/run-20261011T2118Z/run.json

Figure 2: Convergence in multiplicity ℓ. Source `run.json`.



Source: research/lit2026_w10_s5_coppersmith_20261011/reports/run-20261011T2118Z/run.json

Figure 3: Finite lattices, 32-bit curve `a3-fast-32bit-p256class`. Source `run.json`.